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Department of Electrical and Computer Engineering
ECE 371 Smart Grid and Renewable Energy Systems
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Design Project

Design of a Renewable System

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1. OBJECTIVES

Photovoltaic (PV) systems undergo different types of analysis and design work to check for project feasibility. This report aims to investigate the prospects of installing a PV system in a grid-tied residential home using the System Advisor Model (SAM). An accurate model of the PV system is required for accurate energy production data. Economic analysis is reliant on an accurate model of the system cost, energy production, and pricing of the energy markets. Sensitivity analysis is performed to verify the PV system further and to ensure reliability of the proposed system.

2. System Description

The system description will cover the hardware, sizing, and layout used in the design.

2.1 Hardware

The two main components of any PV system are the panel and inverter. There are important design parameters for both components, which must be considered to ensure compatibility between the two.

Table 1 is an organizer that lists the components used in the PV system, paired with the manufacturer and the name of the component. The panel is the SPR-MAX3-415-BLK-R manufactured by Maxeon Solar Technologies. The module is rated as 414.8W at standard test conditions (STC). The open voltage and short circuit current of the module are 40.7V and 12.6A respectively. The inverter is the Solis-1P4K-4G-US manufactured by Ginlong Technologies. The 4kW inverter has a maximum DC voltage of 500V and a maximum DC current of 12.5A.

Table 1: System Hardware

Type	Manufacturer	Component Name
Panel	Maxeon Solar Technologies Ltd.	SPR-MAX3-415-BLK-R
Inverter	Ginlong Technologies Co - Ltd	Solis-1P4K-4G-US

2.2 System Sizing

An accurate model of the load is important for system sizing. The residence on average consumes 0.879kW and peaks at 5.952kW. The total yearly consumption of the residence is 7700kWh. Fig. 1 is a load duration curve for the residential home, where power is on the y-axis and time is on the x-axis. Energy consumption can be found by the area under this curve.

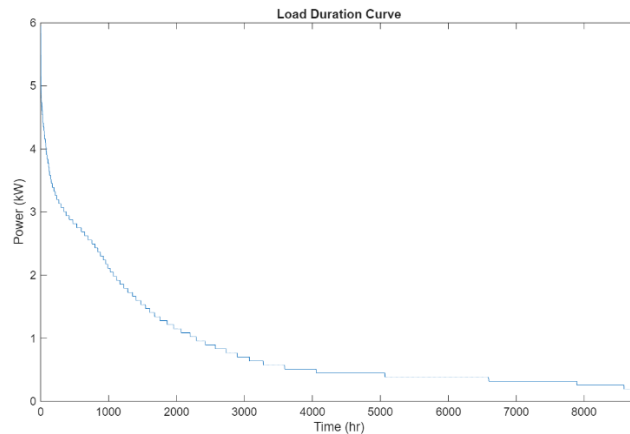


Figure 1: Load duration curve of the residential home

Fig. 2 is a monthly energy consumption chart of the residential home. The energy consumption peaks during the summer months, which is common for residential homes. This is due to an increase usage of air conditioning during the summer months.

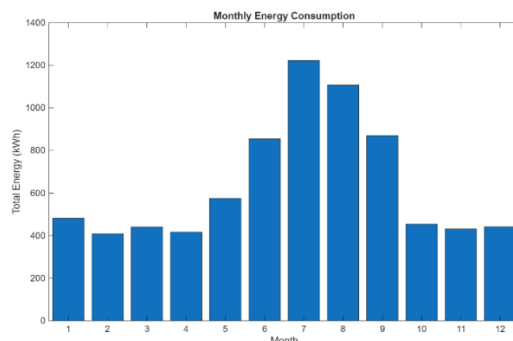


Figure 2: Monthly energy consumption of the residential home

The resource model is obtained in a report from the Open Energy Data Initiative (OEDI), where there is information on global horizontal irradiance (GHI), direct normal irradiance (DNI), diffuse horizontal irradiance (DHI), and temperature. The data is collected in various years starting from 2001 and is within a mile radius of the residence. The resource model from OEDI should be adequate for the system model.

Table 2 is system sizing and energy generation data from the SAM simulation.

Table 2: System sizing and energy generated

Panel DC (kW)	Inverter AC (kW)	Year 1 Energy (kWh)	DC Capacity Factor (%)
4.1	4	6683	18.39

2.3 System Design

The system design must consider the components used, the load, and the interconnections with the grid. Fig. 3 illustrates the layout of the PV modules and the connections made with the inverter and utility meter. The ten PV modules are in series with each other, so the V_{OC} of the PV system is 407V. This is just below the inverter maximum voltage of 500V. There is a single inverter that converts the PVs direct current (DC) to alternating current (AC). This AC feeds into the utility meter with a breaker acting as a switch. The utility meter interfaces with the grid and tracks the importing and exporting of electricity.

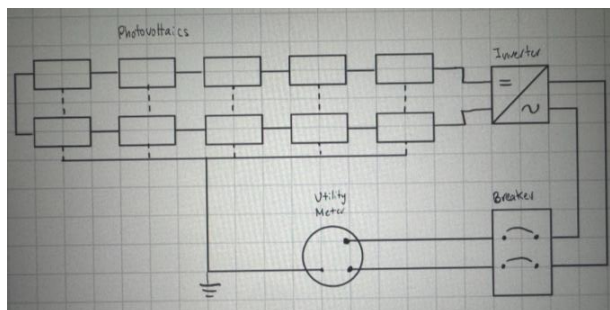


Figure 3: Line diagram of the PV system

3. Sensitivities

This model is heavily reliant on future events. There are many types of metering and billing that utilities enforce, which may change the returns of the PV system. Electricity escalation is not always predictable and can also impact the return of the PV system. Sensitivity analysis can help develop an idea of how different factors impact the system.

Net metering is a system where the PV owners are credited with the energy that is produced. When the system operates with net metering with 0% electricity price escalation, the net present value (NPV) is \$8,148. This will be used as the base case that will be compared to.

Buy all/sell all is a simple billing system, where the sell price is the same for any hour or day. With a sell price of 3 cents/kWh, the NPV is \$1900. With a sell price of 5 cents/kWh, the NPV is \$3571. With a sell price of 7 cents/kWh, the NPV is \$5242. As the sell price increases, the NPV of the system increases. It should also be noted that when the sell price is less than the buy price, buy all/sell all is not as profitable as net metering.

Electricity price escalation at 1% has a NPV of \$8793. Electricity price escalation set at 3% has a NPV of \$10991. Electricity price escalation set at 5% has a NPV of \$13637. As electricity price escalation increases, the NPV of the system increases. So, when the electricity escalation is greater than 0%, the NPV is higher than the base case.

Idaho Power's current tariff is reliant on the time of day and day of the year, also known as net billing. The NPV of the system operating with a net billing utility is \$5359. This is less than net metering, however competitive with buy all/sell all.

3.1 Net Present Value

Table 3 are documented NPV's for each of the billing type. Each simulation assumes a 3% electricity price escalation. The buy all/sell all is set at 5 cents per kWh. For PV system owners, net metering is the best scenario for profitability. Where the residence is located in, the net billing is the most accurate and is the NPV that can be expected.

Table 3: Net Present Value

Net Metering	Buy All/Sell All	Net Billing
\$10991	\$4924	\$7294

3.2 Subsidies and tax credits

There are two tax credits available for the PV system. The Federal Investment Tax Credit (ITC) offers 30% of the system cost as credit.

The Idaho Tax Deduction offers an initial 40% of the system cost as a deduction for taxable income. For the following three years, 20% of the system cost can be deducted further.

3.3 System Cost

According to the Spring 2025 Solar Industry Update from the National Renewable Energy Laboratory (NREL) the average price for residential PV systems in 2024 is \$2.50/W. The default operating cost by SAM, \$28.78/kW-yr, is fairly accurate and used for this model.

PV systems are often paid through a loan as it makes the investment more profitable. A loan for the PV system can be acquired through local credit unions or through Idaho's State Energy Loan Program. The State Energy Loan Program offers different loans, however the 3 year and 3% interest loan is the most profitable for the system.

4. Proposed System

The proposed system for the grid-tied residential home is ten SPR-MAX3-415-BLK-R PV modules paired with the Solis-1P4K-4G-US inverter. The PV modules should be connected in series to be compatible with the inverter.

The tilt and azimuth angles are fixed at 21 and 170 degrees respectively. This is due to the angle of the roof, and that it is difficult to tilt PV modules on residential roofs.

Fig. 4 is a plot of the monthly energy consumption and the monthly energy production of the proposed system.

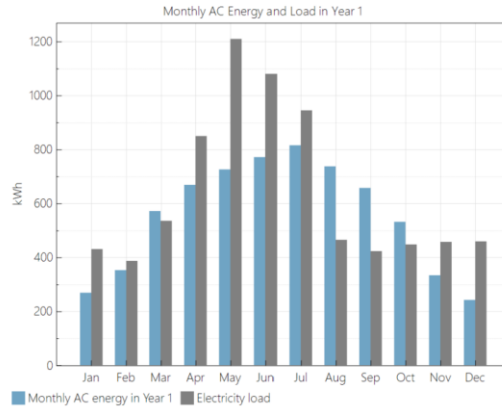


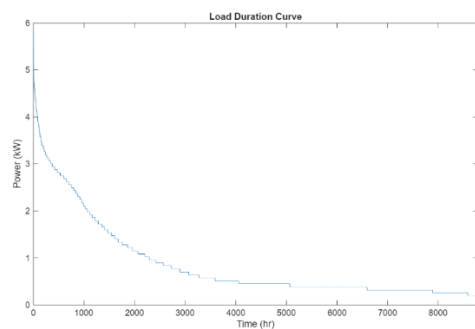
Figure 4: Monthly consumption vs production

The Levelized Cost of Energy (LCOE) calculates the cost of energy. Assuming that all incentives are applied, the loan is 3 years and 3%, and that uses Idaho Power's tariff, the LCOE is 0.90 cents per kWh. The NPV is \$7294 with a discounted payback period of 3.3 years.

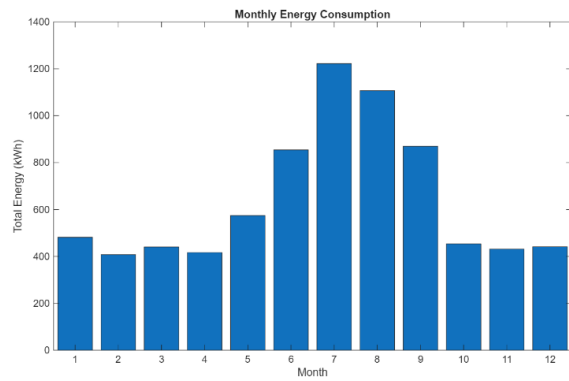
5. CONCLUSION

The proposed system produces 6683 kWh a year with a LCOE of 0.90 cents per kWh. The PV system does economically make sense for the grid-tied residence, which largely comes from federal and state incentives.

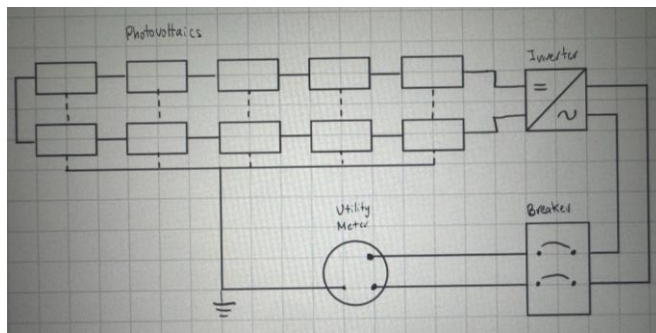
APPENDIX (A)



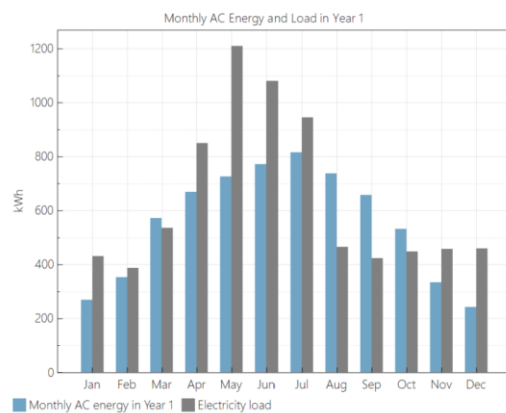
Load duration curve of the residential home



Monthly energy consumption of the residential home



Line diagram of the PV system



Monthly energy consumption vs monthly energy production